Machine learning

Name: Dana Hassan Bafaqih
ID: 445013918
Section number: 2

Dr. Afaf Almehmadi

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1. Data Preprocessing

1.1 Load both datasets.

First dataset (Breast Cancer Wisconsin)

```
# Read the data (Breast Cancer Wisconsin (Diagnostic) Dataset)
Cancer  = pd.read_csv('/content/wdbc[1].data',
                      header = None,      # The data without header
                      delimiter=' *, *',
                      engine='python')
```

```
# Add header to the dataset
Cancer.columns = ['ID','Diagnosis' , 'radius1', 'texture1', 'perimeter1', 'area1',
                  'smoothness1', 'compactness1', 'concavity1',
                  'concave_points1', 'symmetry1', 'fractal_dimension1',
                 'radius2','texture2', 'perimeter2',
                'area2', 'smoothness2', 'compactness2', 'concavity2',
                'concave_points2', 'symmetry2', 'fractal_dimension2', 'radius3',
                'texture3', 'perimeter3', 'area3', 'smoothness3',
                'compactness3', 'concavity3', 'concave_points3','symmetry3',
                'fractal_dimension3']
```

Second dataset (Adult Income)

```
# Read the data (Adult Income Dataset (Census Income))
adult = pd.read_csv('/content/adult[1].data',
                    header = None,              # The data without header
                    delimiter=' *, *', engine='python')
```

```
# Add header to the dataset
adult.columns = ['age', 'workclass', 'fnlwgt', 'education', 'education_num',
                 'marital_status', 'occupation', 'relationship',
                 'race', 'sex', 'capital_gain', 'capital_loss',
                 'hours_per_week', 'native_country', 'income']
```

1.2 Handling Missing Value

First dataset

```
# Show information about the data (columns, type, null count)
Cancer.isnull().sum()
```

	0
ID	0
Diagnosis	0
radius1	0
texture1	0
perimeter1	0
area1	0
smoothness1	0
compactness1	0
concavity1	0
concave_points1	0
symmetry1	0
fractal_dimension1	0
radius2	0
texture2	0
perimeter2	0
area2	0
smoothness2	0
compactness2	0
concavity2	0
concave_points2	0
symmetry2	0
fractal_dimension2	0
radius3	0
texture3	0
perimeter3	0
area3	0
smoothness3	0
compactness3	0
concavity3	0
concave_points3	0
symmetry3	0
fractal_dimension3	0

dtype: int64

There are no missing values in the.

Second dataset

```
# Check for missing vlues value
adult.isnull().sum()
```

age	0
workclass	0
fnlwgt	0
education	0
education_num	0
marital_status	0
occupation	0
relationship	0
race	0
sex	0
capital_gain	0
capital_loss	0
hours_per_week	0
native_country	0
income	0

dtype: int64

Even though it shows no missing value (null), but the data source indicates that some value are missing!

This means they may not be null, but the column contains invalid or missing values.

```
# Check the unique values for columns that may contain missing or invalid values
for col in ['workclass', 'occupation', 'native_country']:
    print(f"Column: {col}")
    print(adult[col].unique())
    print("\n")
```

```
Column: workclass
['State-gov' 'Self-emp-not-inc' 'Private' 'Federal-gov' 'Local-gov' '?'
 'Self-emp-inc' 'Without-pay' 'Never-worked']


Column: occupation
['Adm-clerical' 'Exec-managerial' 'Handlers-cleaners' 'Prof-specialty'
 'Other-service' 'Sales' 'Craft-repair' 'Transport-moving'
 'Farming-fishing' 'Machine-op-inspct' 'Tech-support' '?'
 'Protective-serv' 'Armed-Forces' 'Priv-house-serv']


Column: native_country
['United-States' 'Cuba' 'Jamaica' 'India' '?' 'Mexico' 'South'
 'Puerto-Rico' 'Honduras' 'England' 'Canada' 'Germany' 'Iran'
 'Philippines' 'Italy' 'Poland' 'Columbia' 'Cambodia' 'Thailand' 'Ecuador'
 'Laos' 'Taiwan' 'Haiti' 'Portugal' 'Dominican-Republic' 'El-Salvador'
 'France' 'Guatemala' 'China' 'Japan' 'Yugoslavia' 'Peru'
 'Outlying-US(Guam-USVI-etc)' 'Scotland' 'Trinadad&Tobago' 'Greece'
 'Nicaragua' 'Vietnam' 'Hong' 'Ireland' 'Hungary' 'Holand-Netherlands']
```

We can observe that some entries contain '**?**' value

```
# Replace the '?' with the mode because these are categorical features
for col in ['workclass', 'occupation', 'native_country']:
  adult[col] = adult[col].replace('?', adult[col].mode()[0])
```

```
# Check again
for col in ['workclass', 'occupation', 'native_country']:
    print(f"Column: {col}")
    print(adult[col].unique())
    print("\n")
```

```
Column: workclass
['State-gov' 'Self-emp-not-inc' 'Private' 'Federal-gov' 'Local-gov'
 'Self-emp-inc' 'Without-pay' 'Never-worked']


Column: occupation
['Adm-clerical' 'Exec-managerial' 'Handlers-cleaners' 'Prof-specialty'
 'Other-service' 'Sales' 'Craft-repair' 'Transport-moving'
 'Farming-fishing' 'Machine-op-inspct' 'Tech-support' 'Protective-serv'
 'Armed-Forces' 'Priv-house-serv']


Column: native_country
['United-States' 'Cuba' 'Jamaica' 'India' 'Mexico' 'South' 'Puerto-Rico'
 'Honduras' 'England' 'Canada' 'Germany' 'Iran' 'Philippines' 'Italy'
 'Poland' 'Columbia' 'Cambodia' 'Thailand' 'Ecuador' 'Laos' 'Taiwan'
 'Haiti' 'Portugal' 'Dominican-Republic' 'El-Salvador' 'France'
```

1.3 Data Encoding

I apply encoding to convert the categorical columns into numbers. I also apply it to the target column in both datasets to make it easier to work with.

First dataset

```
# Encode the 'Diagnosis' column with LabelEncoder because it's binary
le = LabelEncoder()
Cancer['Diagnosis'] = le.fit_transform(Cancer['Diagnosis'])

print(Cancer.head())
```

```
         ID  Diagnosis  radius1  texture1  perimeter1   area1  smoothness1  \
0    842302          1    17.99     10.38      122.80  1001.0      0.11840
1    842517          1    20.57     17.77      132.90  1326.0      0.08474
2  84300903          1    19.69     21.25      130.00  1203.0      0.10960
3  84348301          1    11.42     20.38       77.58   386.1      0.14250
4  84358402          1    20.29     14.34      135.10  1297.0      0.10030

   compactness1  concavity1  concave_points1  ...  radius3  texture3  \
0       0.27760      0.3001          0.14710  ...    25.38     17.33
1       0.07864      0.0869          0.07017  ...    24.99     23.41
2       0.15990      0.1974          0.12790  ...    23.57     25.53
3       0.28390      0.2414          0.10520  ...    14.91     26.50
4       0.13280      0.1980          0.10430  ...    22.54     16.67

   perimeter3   area3  smoothness3  compactness3  concavity3  concave_points3  \
0      184.60  2019.0       0.1622        0.6656      0.7119           0.2654
1      158.80  1956.0       0.1238        0.1866      0.2416           0.1860
2      152.50  1709.0       0.1444        0.4245      0.4504           0.2430
3       98.87   567.7       0.2098        0.8663      0.6869           0.2575
4      152.20  1575.0       0.1374        0.2050      0.4000           0.1625

   svmmetrv3  fractal dimension3
```

Second dataset

```
# Encode the 'sex' and 'income' columns with LabelEncoder because it's binary
le = LabelEncoder()
for col in ['sex', 'income']:
    adult[col] = le.fit_transform(adult[col])

# Encode remaining categorical features with One-Hot because there is no ordering in the data
adultE = pd.get_dummies( adult, columns=['workclass', 'education', 'marital_status', 'occupation',
             'relationship', 'race', 'native_country'], dtype=int)

print(adultE.head())
```

```
   age  fnlwgt  education_num  sex  capital_gain  capital_loss  \
0   39   77516             13    1          2174             0
1   50   83311             13    1             0             0
2   38  215646              9    1             0             0
3   53  234721              7    1             0             0
4   28  338409             13    0             0             0

   hours_per_week  income  workclass_Federal-gov  workclass_Local-gov  ...  \
0              40       0                      0                    0  ...
1              13       0                      0                    0  ...
2              40       0                      0                    0  ...
3              40       0                      0                    0  ...
4              40       0                      0                    0  ...

   native_country_Portugal  native_country_Puerto-Rico  \
0                        0                           0
```

1.4 Data Standardization

I apply standardization to numerical features so all columns have the same scale. If it is not used, features with large values can dominate the calculations, causing bias and lower accuracy for SVM and clustering.

First dataset

```
# Split features and target
CancerX = Cancer.drop(['Diagnosis'], axis=1)  # Feature
Cancery = Cancer['Diagnosis']                 # Target
```

```
# Applying Standardization to scale features
scaler = StandardScaler()
CancerX_stand = scaler.fit_transform(CancerX)
```

Second dataset

```
# Split features and target
adultX = adultE.drop(['income'], axis=1)   # Feature
adulty = adultE['income']                  # Target
```

```
# Applying Standardization to scale features
scaler = StandardScaler()
adultX_stand = scaler.fit_transform(adultX)
```

1.5 Split the Dataset (for the supervised)

First dataset

```
#Split the dataset into training and testing sets (70% train, 30% test).
Xc_train, Xc_test, yc_train, yc_test = train_test_split(
    CancerX_stand, Cancery, test_size=0.3, random_state=42
    )
```

Second dataset

```
#Split the dataset into training and testing sets (70% train, 30% test).
Xa_train, Xa_test, ya_train, ya_test = train_test_split(
    adultX_stand, adulty, test_size=0.3, random_state=42
    )
```

2. Model Building

2.1 Supervised Model: SVM

I used two Kernel:

- **Linear Kernel:** is used when the data is *linearly separable*. It separates the data using a single straight line. It has one hyperparameter **C** which balances between maximizing the margin and penalizing misclassified points. A larger **C** means a stricter penalty for errors, while a smaller **C** allows a wider margin and more tolerance for misclassification.
- **RBF Kernel:** is used for *non-linear data* It transforms the data into a higher-dimensional space to create a non-linear decision boundary. It has two hyperparameters: **C** (same as in Linear Kernel) and **gamma**, which determines how far the influence of each training point reaches.

- I used **GridSearchCV** on both datasets to find the **best hyperparameters** and show the best result

First Dataset

```
#Linear Kernel SVM
linear_paramsC = {
    "C": [0.1, 1, 10, 50, 100]
}

linear_svmC = SVC(kernel="linear", probability=True)

grid_linearC = GridSearchCV(linear_svmC, linear_paramsC, cv=5, scoring="accuracy")
grid_linearC.fit(Xc_train, yc_train)

print("Best Linear SVM params:", grid_linearC.best_params_)
print("Best Linear SVM score:", grid_linearC.best_score_)
```

```
Best Linear SVM params: {'C': 0.1}
Best Linear SVM score: 0.9748417721518987
```

```
#RBF Kernel SVM
rbf_paramsC = {
   "C": [0.1, 1, 10, 50, 100],
   "gamma": ["scale", 0.1, 0.01, 0.001]
}

rbf_svmC = SVC(kernel="rbf", probability=True)

grid_rbfC = GridSearchCV(rbf_svmC, rbf_paramsC, cv=5, scoring="accuracy")
grid_rbfC.fit(Xc_train, yc_train)

print("Best RBF params:", grid_rbfC.best_params_)
print("Best RBF score:", grid_rbfC.best_score_)
```

```
Best RBF params: {'C': 50, 'gamma': 0.001}
Best RBF score: 0.9723101265822784
```

Second Dataset

The model took a VERY long time to train on the full dataset and gave an error instead of output. So, I trained it on a random sample.

```
# Take a random sample (20%) from the training data to speed up training
sample_frac = 0.2
Xa_train_df = pd.DataFrame(Xa_train)  # Convert training features to DataFrame
Xa_train_df['target'] = ya_train.values  # Temporarily add the target column

# Sample the data
Xa_train_sample = Xa_train_df.sample(frac=sample_frac, random_state=42)

# Separate features and target again
ya_train_sample = Xa_train_sample['target']
Xa_train_sample = Xa_train_sample.drop('target', axis=1)
```

```
# Linear SVM
linear_svmA = SVC(kernel='linear', probability=True)

# Define hyperparameters for GridSearchCV
linear_paramsA = {"C":  [0.1, 1, 10, 50]}

# GridSearchCV for hyperparameter tuning
grid_linearA = GridSearchCV(linear_svmA, linear_paramsA, cv=3, scoring='accuracy')
grid_linearA.fit(Xa_train_sample, ya_train_sample)

print("Best Linear SVM best params:", grid_linearA.best_params_)
print("Best Linear SVM score:", grid_linearA.best_score_)
```

```
Best Linear SVM best params: {'C': 1}
Best Linear SVM score: 0.840503476433018
```

```
# RBF SVM
rbf_svmA = SVC(kernel='rbf', probability=True)

# Hyperparameters for GridSearchCV
rbf_paramsA = {
   "C": [0.1, 1, 10, 50, 100],
   "gamma": ["scale", 0.1, 0.01, 0.001]
}

# GridSearchCV for hyperparameter tuning
grid_rbfA = GridSearchCV(rbf_svmA, rbf_paramsA, cv=3, scoring='accuracy')
grid_rbfA.fit(Xa_train_sample, ya_train_sample)

print("Best Linear SVM best params:", grid_rbfA.best_params_)
print("Best Linear SVM score:", grid_rbfA.best_score_)
```

```
Best Linear SVM best params: {'C': 100, 'gamma': 0.001}
Best Linear SVM score: 0.8418196990171282
```

It still takes time because of the **probability=True** parameter, which I need It to calculate the AUC score, but it is faster than training on the whole dataset.

* *Larger **C** and **gamma** make the model stricter and the margin smaller, while smaller values allow a wider margin and smoother decision boundary.*

2.2 Unsupervised Models: K-Means & Hierarchical Clustering

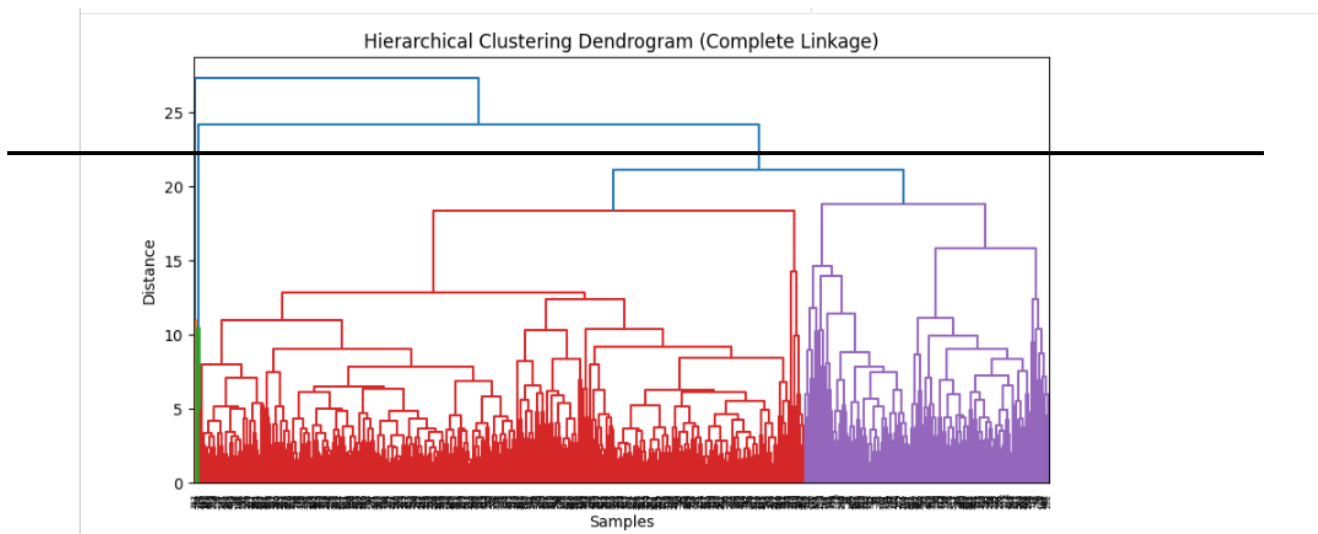
First Dataset

- **K-Means**

```
# K-Means Clustering
kmeansC = KMeans(n_clusters=2, random_state=42) #n_clusters=2 because there are two classes in the dataset.
kmeansC.fit(CancerX_stand) #only features, no labels

kmeans_labelsC = kmeansC.fit_predict(CancerX_stand)
```

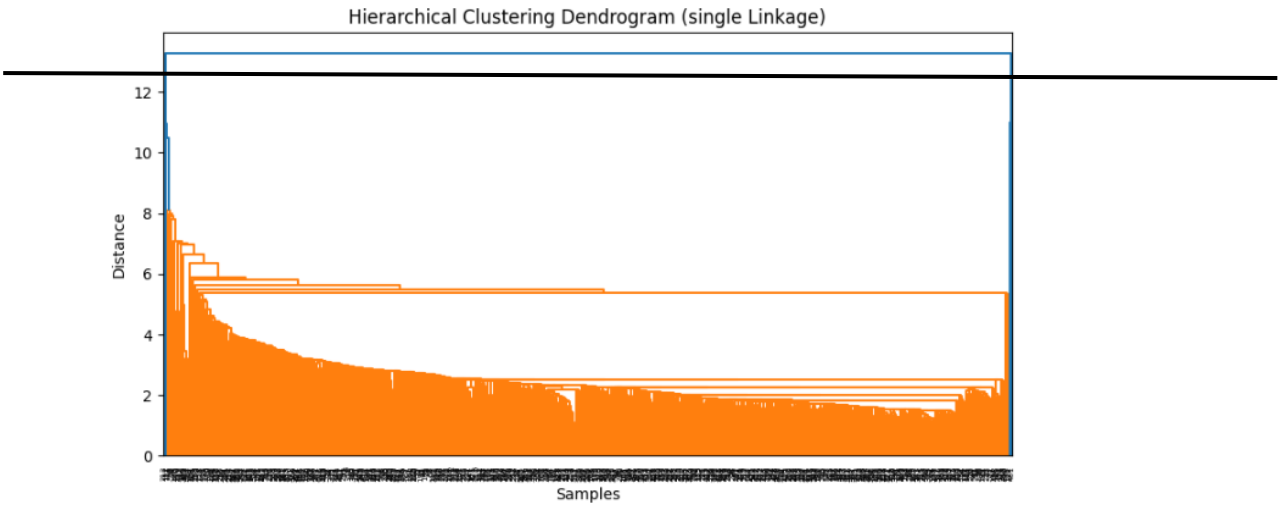
- **Hierarchical Clustering**

```
# Compute linkage (Hierarchical Clustering)
completeC = linkage(CancerX_stand, method='complete')

#Plots dendrogram (Agglomerative)
plt.figure(figsize=(10, 5))
dendrogram(completeC)
plt.title("Hierarchical Clustering Dendrogram (Complete Linkage)")
plt.xlabel("Samples")
plt.ylabel("Distance")
plt.show()

#cut the dendrogram to get a specific number of clusters using fcluster
clustersC = fcluster(completeC, t=2, criterion='maxclust')  #t=2: Specifies the desired number of clusters.
```



There is a large distance between 20 and 25 where no merges occur. So, when I cut the dendrogram at this height, I draw a horizontal line and count how many vertical lines it crosses, which gives **2 clusters**

```python
# Compute linkage (Hierarchical Clustering)
singleC = linkage(CancerX_stand, method='single')

#Plots dendrogram (Agglomerative)
plt.figure(figsize=(10, 5))
dendrogram(singleC)
plt.title("Hierarchical Clustering Dendrogram (single Linkage)")
plt.xlabel("Samples")
plt.ylabel("Distance")
plt.show()

#cut the dendrogram to get a specific number of clusters using fcluster
clustersC2 = fcluster(singleC, t=2, criterion='maxclust')  #t=2: Specifies the desired number of clusters.
```



* *In short, I looked for the largest horizontal gaps in the dendrogram, and by cutting it at that point, I obtained the number of clusters that best fits the data, which is **2 clusters**.*

Second Dataset

- K-Means

```python
kmeansA = KMeans(n_clusters=2, random_state=42)
kmeansA.fit(adultX_stand) #only features, no labels

kmeans_labelsA = kmeansA.fit_predict(adultX_stand)
```

- Hierarchical Clustering

```python
# Compute linkage (Hierarchical Clustering)
completeA = linkage(adultX_stand, method='complete')

#cut the dendrogram to get a specific number of clusters using fcluster
clustersA = fcluster(completeA, t=2, criterion='maxclust')  #t=2: Specifies the desired number of clusters.
```

```python
# Compute linkage (Hierarchical Clustering)
singleA = linkage(adultX_stand, method='single')

#cut the dendrogram to get a specific number of clusters using fcluster
clustersA2 = fcluster(singleA, t=2, criterion='maxclust')  #t=2: Specifies the desired number of clusters.
```

3. Model Evaluation and Visualization

3.1 SVM Evaluation (Supervised)

3.1.1 Predict class labels

First Dataset

```
#Predict class labels
print("\nSVM With Linear Kernel:")
y_pred_linearC = grid_linearC.predict(Xc_test)
print(y_pred_linearC)

print('\n')

print("\nSVM With RBF Kernel:")
y_pred_rbfC = grid_rbfC.predict(Xc_test)
print(y_pred_rbfC)
```

```
SVM With Linear Kernel:
[0 1 1 0 0 1 1 1 0 0 0 1 0 1 0 1 0 0 0 1 0 0 1 0 0 0 0 0 0 1 0 0 0 0 0 0 1
 0 1 0 0 1 0 0 0 0 0 0 0 0 1 1 0 0 0 0 0 1 1 0 0 1 1 0 0 0 1 1 0 0 1 1 0 1
 0 0 0 0 0 0 1 0 1 1 1 1 1 1 0 0 0 0 0 0 0 0 1 1 0 1 1 0 1 1 0 0 0 1 0 0 1
 0 0 1 1 1 0 0 0 1 0 0 0 1 0 1 1 0 0 1 1 1 0 0 0 1 0 0 0 1 0 1 0 0 1 0 1 1
 1 0 1 0 0 0 0 1 1 0 0 0 0 0 0 0 1 0 0 0 0 1 0]


SVM With RBF Kernel:
[0 1 1 0 0 1 1 1 0 0 0 1 0 1 0 1 0 0 0 1 0 0 1 0 0 0 0 0 0 1 0 0 0 0 0 0 1
 0 1 0 0 1 0 0 0 0 0 0 0 0 1 1 0 0 0 0 0 1 1 0 0 1 1 0 0 0 1 1 0 0 1 1 0 1
 0 0 0 0 0 0 1 0 1 1 1 1 1 1 0 0 0 0 0 0 0 0 1 1 0 1 1 0 1 1 0 0 0 1 0 0 1
 0 0 1 1 1 0 0 0 1 0 0 0 1 0 1 1 0 0 1 1 1 0 0 0 1 0 0 0 1 0 1 0 0 1 0 1 1
 1 0 1 0 0 0 0 1 1 0 0 0 0 0 0 0 1 0 0 0 0 1 0]
```

Second Dataset

```
print("\nSVM With Linear Kernel:")
y_pred_linearA = grid_linearA.predict(Xa_test)
print(y_pred_linearA)

print('\n')

print("\nSVM With RBF Kernel:")
y_pred_rbfA = grid_rbfA.predict(Xa_test)
print(y_pred_rbfA)
```

```
SVM With Linear Kernel:
[0 0 1 ... 0 0 1]


SVM With RBF Kernel:
[0 0 1 ... 0 0 1]
```

3.1.2 Evaluation Metrics

First Dataset

```python
print("\nSVM With Linear Kernel:")
# Model Evaluation Metrics
print("\nAccuracy:", accuracy_score(yc_test, y_pred_linearC))
print("\nPrecision:", precision_score(yc_test, y_pred_linearC))
print("\nRecall:", recall_score(yc_test, y_pred_linearC))
print("\nF1 Score:", f1_score(yc_test, y_pred_linearC))
print("\nAUC - ROC Score:", roc_auc_score(yc_test, grid_linearC.predict_proba(Xc_test)[:,1]))

print("\n----------------------------------------------")

print("\nSVM With RBF Kernel")
print("\nAccuracy:", accuracy_score(yc_test, y_pred_rbfC))
print("\nPrecision:", precision_score(yc_test, y_pred_rbfC))
print("\nRecall:", recall_score(yc_test, y_pred_rbfC))
print("\nF1 Score:", f1_score(yc_test, y_pred_rbfC))
print("\nAUC - ROC Score:", roc_auc_score(yc_test, grid_rbfC.predict_proba(Xc_test)[:,1]))
```

```
SVM With Linear Kernel:

Accuracy: 0.9824561403508771

Precision: 0.9838709677419355

Recall: 0.9682539682539683

F1 Score: 0.976

AUC - ROC Score: 0.9976484420928865

----------------------------------------------

SVM With RBF Kernel

Accuracy: 0.9824561403508771

Precision: 0.9838709677419355

Recall: 0.9682539682539683

F1 Score: 0.976

AUC - ROC Score: 0.9979423868312758
```

It gives the **same** result, which may be because the data is linearly separable. The Linear SVM already finds the optimal decision boundary, so the RBF kernel does not provide any additional improvement.

Second Dataset

```
print("\nSVM With Linear Kernel:")
# Model Evaluation Metrics
print("\nAccuracy:", accuracy_score(ya_test, y_pred_linearA))
print("\nPrecision:", precision_score(ya_test, y_pred_linearA))
print("\nRecall:", recall_score(ya_test, y_pred_linearA))
print("\nF1 Score:", f1_score(ya_test, y_pred_linearA))
print("\nAUC - ROC Score:", roc_auc_score(ya_test, grid_linearA.predict_proba(Xa_test)[:,1]))

print("\n----------------------------------------------")

print("\nSVM With RBF Kernel")
print("\nAccuracy:", accuracy_score(ya_test, y_pred_rbfA))
print("\nPrecision:", precision_score(ya_test, y_pred_rbfA))
print("\nRecall:", recall_score(ya_test, y_pred_rbfA))
print("\nF1 Score:", f1_score(ya_test, y_pred_rbfA))
print("\nAUC - ROC Score:", roc_auc_score(ya_test, grid_rbfA.predict_proba(Xa_test)[:,1]))
```

```
SVM With Linear Kernel:

Accuracy: 0.846043607329307

Precision: 0.7113778705636743

Recall: 0.5890233362143474

F1 Score: 0.6444444444444445

AUC - ROC Score: 0.8952746151353526

----------------------------------------------

SVM With RBF Kernel

Accuracy: 0.8444057733647251

Precision: 0.7122994652406417

Recall: 0.5756266205704408

F1 Score: 0.6367112810707457

AUC - ROC Score: 0.8879218845194474
```

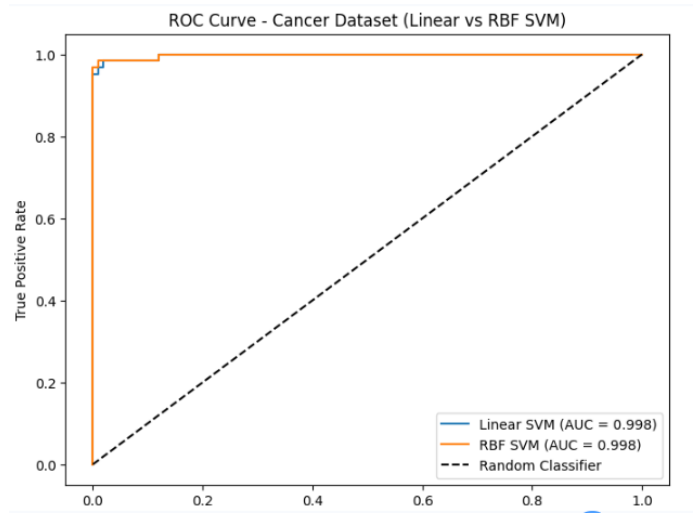
Both Linear and RBF SVM performed **similarly**. The Linear SVM has slightly higher accuracy, recall, F1 score, and AUC, while the RBF SVM has slightly higher precision.

3.1.3 ROC Curves

The closer the ROC curve is to **1**, the **better** the model performs

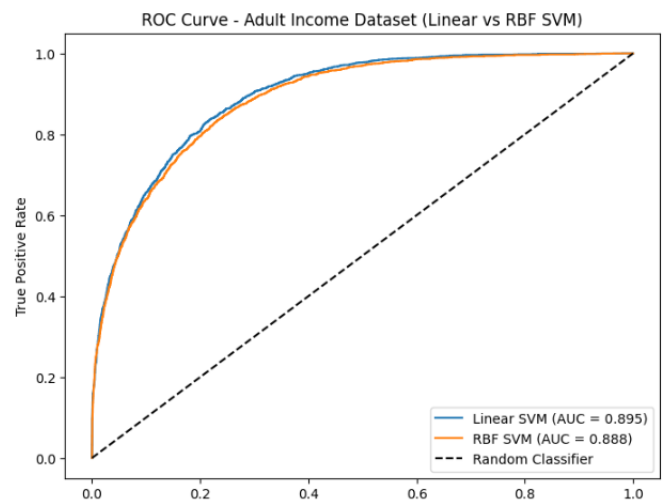
First Dataset

```python
# Compute ROC curves
auc_linear = roc_auc_score(yc_test, grid_linearC.predict_proba(Xc_test)[:,1])
auc_rbf    = roc_auc_score(yc_test, grid_rbfC.predict_proba(Xc_test)[:,1])
fpr_linear, tpr_linear, _ = roc_curve(yc_test, grid_linearC.predict_proba(Xc_test)[:,1])
fpr_rbf, tpr_rbf, _ = roc_curve(yc_test, grid_rbfC.predict_proba(Xc_test)[:,1])

# Plot ROC curves overlayed
plt.figure(figsize=(8,6))
plt.plot(fpr_linear, tpr_linear, label=f"Linear SVM (AUC = {auc_linear:.3f})")
plt.plot(fpr_rbf, tpr_rbf, label=f"RBF SVM (AUC = {auc_rbf:.3f})")
plt.plot([0,1], [0,1], 'k--', label="Random Classifier")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve - Cancer Dataset (Linear vs RBF SVM)")
plt.legend()
plt.show()
```



Second Dataset

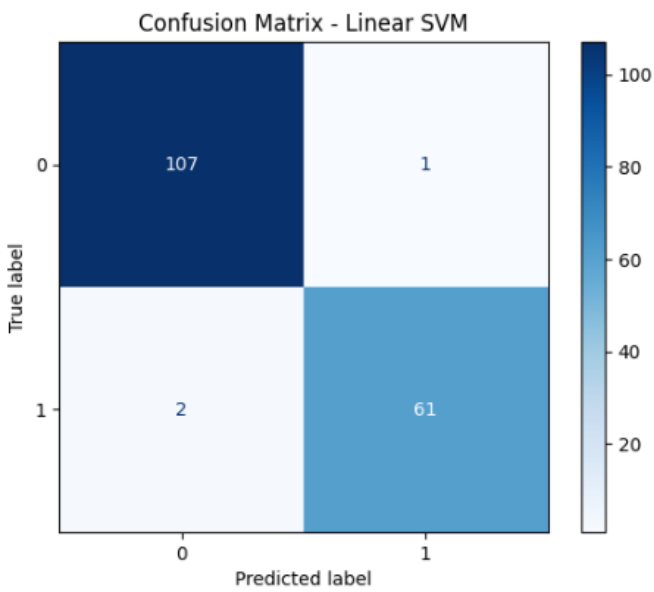
```python
# Compute ROC curves
auc_linearA = roc_auc_score(ya_test, grid_linearA.predict_proba(Xa_test)[:,1])
auc_rbfA    = roc_auc_score(ya_test, grid_rbfA.predict_proba(Xa_test)[:,1])
fpr_linearA, tpr_linearA, _ = roc_curve(ya_test, grid_linearA.predict_proba(Xa_test)[:,1])
fpr_rbfA, tpr_rbfA, _ = roc_curve(ya_test, grid_rbfA.predict_proba(Xa_test)[:,1])

# Plot ROC curves overlayed
plt.figure(figsize=(8,6))
plt.plot(fpr_linearA, tpr_linearA, label=f"Linear SVM (AUC = {auc_linearA:.3f})")
plt.plot(fpr_rbfA, tpr_rbfA, label=f"RBF SVM (AUC = {auc_rbfA:.3f})")
plt.plot([0,1], [0,1], 'k--', label="Random Classifier")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve - Adult Income Dataset (Linear vs RBF SVM)")
plt.legend()
plt.show()
```

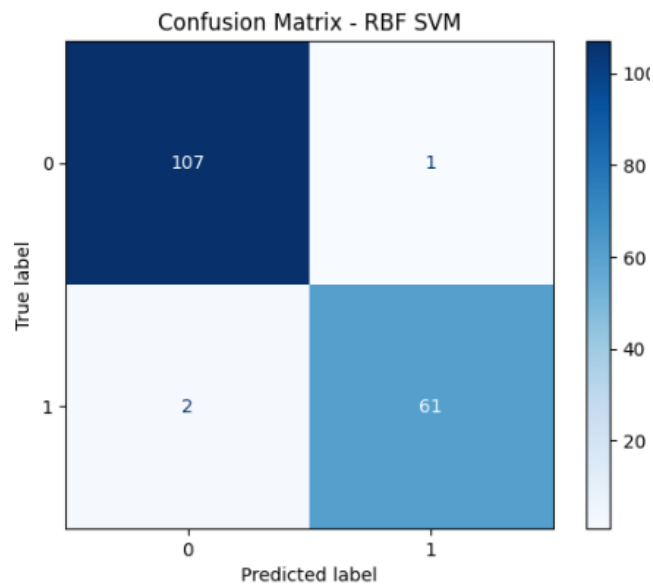


3.1.4 Confusion Matrix

First Dataset

```
# Linear SVM
cm_linear = confusion_matrix(yc_test, y_pred_linearC)
disp_linear = ConfusionMatrixDisplay(confusion_matrix=cm_linear, display_labels=grid_linearC.classes_)
disp_linear.plot(cmap=plt.cm.Blues)
plt.title("Confusion Matrix - Linear SVM")
plt.show()

# RBF SVM
cm_rbf = confusion_matrix(yc_test, y_pred_rbfC)
disp_rbf = ConfusionMatrixDisplay(confusion_matrix=cm_rbf, display_labels=grid_rbfC.classes_)
disp_rbf.plot(cmap=plt.cm.Blues)
plt.title("Confusion Matrix - RBF SVM")
plt.show()
```



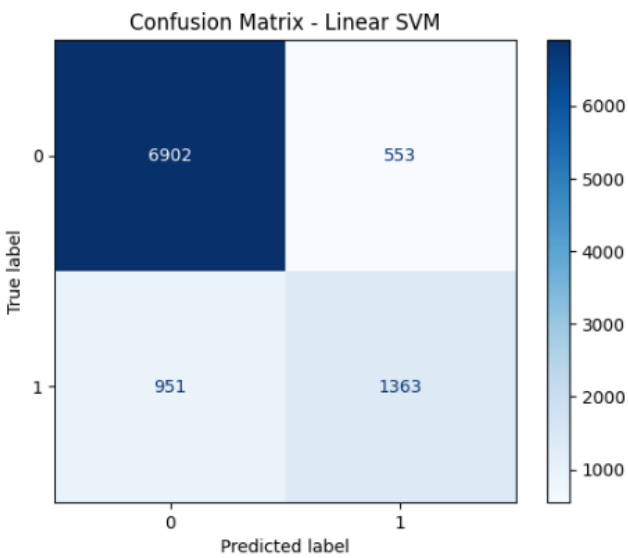


107 = True Positive
The model predicted **1**, and the true label was **1**.
1 = False Negative
The model predicted **0**, but the true label was **1**.
2 = False Positive
The model predicted **1**, but the true label was **0**.
61 = True Negative
The model predicted **0**, and the true label was **0**.

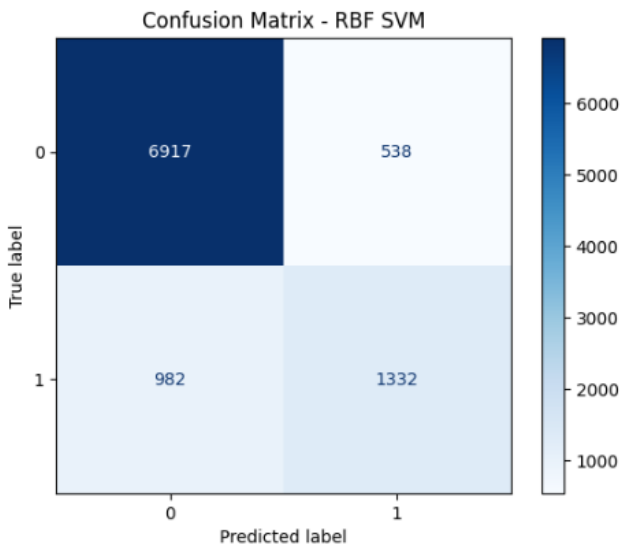
Both models produced the **same** results, and the performance was good because the mistakes were very few.

Second Dataset

```
# Linear SVM
cm_linearA = confusion_matrix(ya_test, y_pred_linearA)
disp_linearA = ConfusionMatrixDisplay(confusion_matrix=cm_linearA, display_labels=grid_linearA.classes_)
disp_linearA.plot(cmap=plt.cm.Blues)
plt.title("Confusion Matrix - Linear SVM")
plt.show()

# RBF SVM
cm_rbfA = confusion_matrix(ya_test, y_pred_rbfA)
disp_rbfA = ConfusionMatrixDisplay(confusion_matrix=cm_rbfA, display_labels=grid_rbfA.classes_)
disp_rbfA.plot(cmap=plt.cm.Blues)
plt.title("Confusion Matrix - RBF SVM")
plt.show()
```



6902 = True Positive
The model predicted 1, and the true label was 1.
553 = False Negative
The model predicted 0, but the true label was 1.
951 = False Positive
The model predicted 1, but the true label was 0.
1363 = True Negative
The model predicted 0, and the true label was 0.



6917 = True Positive
The model predicted 1, and the true label was 1.
538 = False Negative
The model predicted 0, but the true label was 1.
982 = False Positive
The model predicted 1, but the true label was 0.
1332 = True Negative
The model predicted 0, and the true label was 0.

Both models give similar results with only small differences. The **Linear SVM** is better at avoiding false alarms (lower FP, higher TN), while the **RBF SVM** is better at detecting positive cases (higher TP, lower FN).

3.2 Clustering Evaluation (Unsupervised)

3.2.1 Silhouette Score

Silhouette Score measures how similar an object is to its own cluster
The score **ranges** from **-1 to 1**:

- **Close to 1:** Each point is close to its own cluster and far from other clusters **(excellent clustering)**
- **Close to 0:** Points are near the boundaries between clusters
- **Negative (-1):** Some points assigned to the wrong cluster **(poor clustering)**

First Dataset

```
# Silhouette score
#K-Means Clustering
sil_kmeansC = silhouette_score(CancerX_stand, kmeans_labelsC)
print("Silhouette Score (K-Means):", sil_kmeansC)

#Hierarchical Clustering(Complete)
sil_hierC = silhouette_score(CancerX_stand, clustersC)
print("\nSilhouette Score (Hierarchical-Complete):", sil_hierC)

#Hierarchical Clustering(Single)
sil_hierC2 = silhouette_score(CancerX_stand, clustersC2)
print("\nSilhouette Score (Hierarchical-Single):", sil_hierC2)
```

```
Silhouette Score (K-Means): 0.3382381691599185

Silhouette Score (Hierarchical): 0.6655009522688491

Silhouette Score (Hierarchical): 0.6364000118189117
```

(K-Means):
0.338 → near cluster boundaries
(Hierarchical-Complete): 0.665 →points are correctly groped
(Hierarchical-Single): 0.636 → points are correctly groped

Second Dataset

```
# Silhouette Score (K-mean)
sil_kmeansA = silhouette_score(adultX_stand, labels_kmeansA)
print("Silhouette Score - KMeans:", sil_kmeansA)

# Silhouette Score (Hierarchical Clustering)
sil_hierA = silhouette_score(adultX_stand, clustersA)
print("Silhouette Score - Hierarchical (Complete):", sil_hierA)

# Silhouette Score (Hierarchical Clustering)
sil_hierA2 = silhouette_score(adultX_stand, clustersA2)
print("Silhouette Score - Hierarchical (Single):", sil_hierA2)
```

```
Silhouette Score - KMeans: 0.058714951049135475
Silhouette Score - Hierarchical (Complete): 0.9307286383739864
Silhouette Score - Hierarchical (Single): 0.9307286383739864
```

(K-Means): 0.058 → points are correctly groped
(Hierarchical-Complete): 0.9307→ points are correctly groped
(Hierarchical-Single): 0.9307→ points are correctly groped

3.2.2 Clusters VS True Labels

First Dataset

```
#clusters vs true labels
kmeans_tableC = pd.crosstab(kmeans_labelsC, Cancery)
print("\nK-Means Clusters vs True Labels:\n", kmeans_tableC)

hier_tableC = pd.crosstab(clustersC, Cancery)
print("\nHierarchical Clusters(Complete) vs True Labels:\n", hier_tableC)

hier_tableC2 = pd.crosstab(clustersC2, Cancery)
print("\nHierarchical Clusters(Single) vs True Labels:\n", hier_tableC2)
```

```
K-Means Clusters vs True Labels:
 Diagnosis    0    1
row_0
0           13  175
1          344   37

Hierarchical Clusters(Complete) vs True Labels:
 Diagnosis    0    1
row_0
1            0    2
2          357  210

Hierarchical Clusters(Single) vs True Labels:
 Diagnosis    0    1
row_0
1            0    3
2          357  209
```

- **K-Means:** Clusters are mixed, not very clear.
- **Hierarchical (Complete and Single Linkage):** Clusters mostly good, one small cluster

Second Dataset

```
#clusters vs true labels
print("\nKMeans vs True Labels:")
print(pd.crosstab(labels_kmeansA, adulty))

print("\nHierarchical vs True Labels:")
print(pd.crosstab(clustersA, adulty))

print("\nHierarchical vs True Labels:")
print(pd.crosstab(clustersA2, adulty))
```

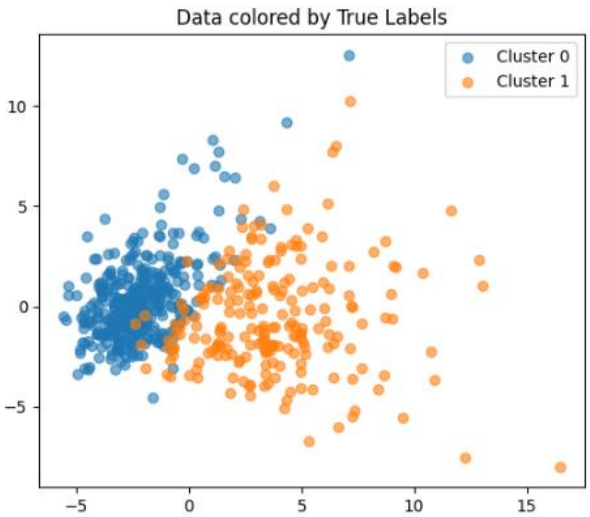
```
KMeans vs True Labels:
income      0     1
row_0
0        8160  6687
1       16560  1154

Hierarchical vs True Labels:
income      0     1
row_0
1       24719  7841
2           1     0

Hierarchical vs True Labels:
income      0     1
row_0
1       24719  7841
2           1     0
```

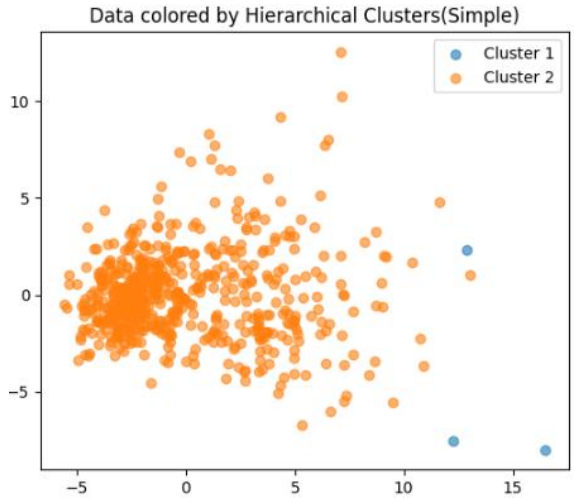
- **K-Means:** Clusters very mixed, bad separation
- **Hierarchical (Complete and Single Linkage):** Clusters mostly good, one small cluster

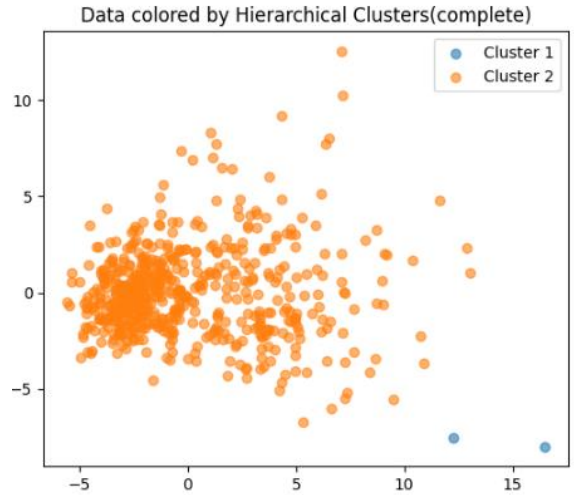
3.2.3 PCA to 2D

First Dataset

```
pca = PCA(n_components=2)
Xa_2d = pca.fit_transform(CancerX_stand)

# Function to plot clusters
def plot_clusters(X_2d, labels, title):
    plt.figure(figsize=(6,5))
    for lab in np.unique(labels):
        plt.scatter(X_2d[labels==lab,0], X_2d[labels==lab,1], label=f'Cluster {lab}', alpha=0.6)
    plt.title(title)
    plt.legend()
    plt.show()

# Plot by true labels
plot_clusters(Xa_2d, Cancery.values, "Data colored by True Labels")

# Plot by KMeans clusters
plot_clusters(Xa_2d, kmeans_labelsC, "Data colored by KMeans Clusters")

# Plot by Hierarchical clusters(complete)
plot_clusters(Xa_2d, clustersC, "Data colored by Hierarchical Clusters(complete)")

# Plot by Hierarchical clusters(Simple)
plot_clusters(Xa_2d, clustersC2, "Data colored by Hierarchical Clusters(Simple)")
```

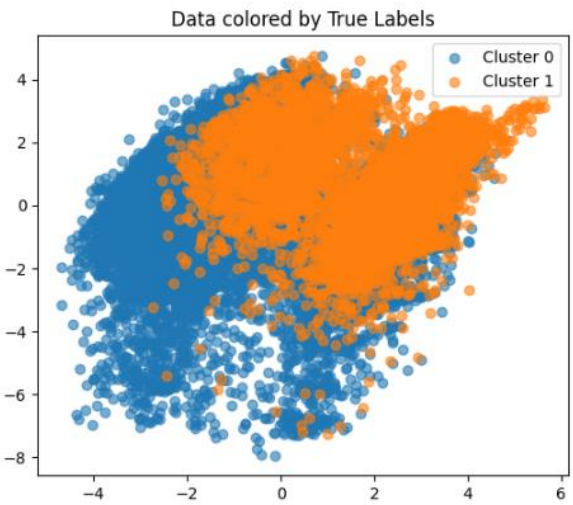


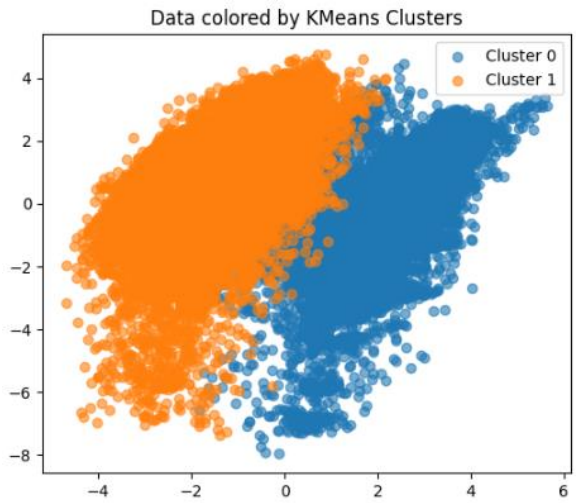


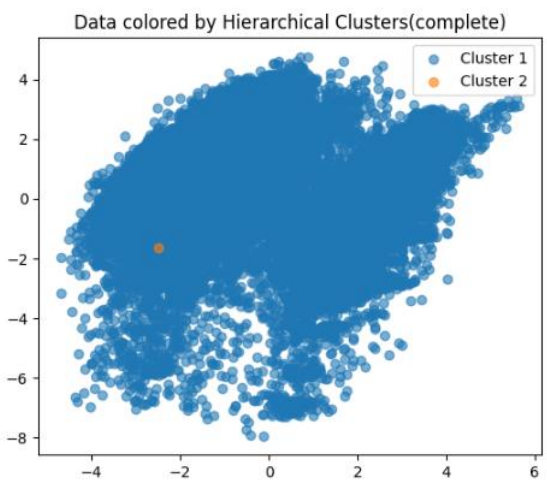


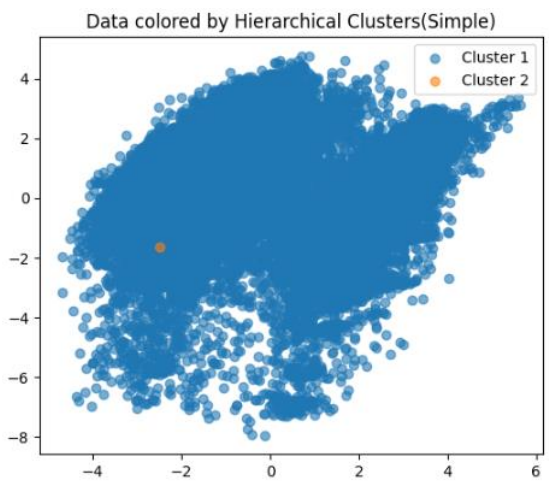
Second Dataset

```
pca = PCA(n_components=2)
Xa_2d = pca.fit_transform(adultX_stand)

# Function to plot clusters
def plot_clusters(X_2d, labels, title):
    plt.figure(figsize=(6,5))
    for lab in np.unique(labels):
        plt.scatter(X_2d[labels==lab,0], X_2d[labels==lab,1], label=f'Cluster {lab}', alpha=0.6)
    plt.title(title)
    plt.legend()
    plt.show()

# Plot by true labels
plot_clusters(Xa_2d, adulty.values, "Data colored by True Labels")

# Plot by KMeans clusters
plot_clusters(Xa_2d, labels_kmeansA, "Data colored by KMeans Clusters")

# Plot by Hierarchical clusters(complete)
plot_clusters(Xa_2d, clustersA, "Data colored by Hierarchical Clusters(complete)")

# Plot by Hierarchical clusters(Simple)
plot_clusters(Xa_2d, clustersA2, "Data colored by Hierarchical Clusters(Simple)")
```









4. Comparative Analysis

1. How does **SVM** performance compare to **K-Means** and **Hierarchical** on each dataset?
   In the first dataset, both SVM models (Linear and RBF kernels) achieve significantly better performance, with all evaluation scores in the 90s.
   K-Means does not perform well, it has a low Silhouette Score of 0.33, which is close to 0, indicating that the clusters are poorly separated and very sensitive to small changes. The clusters-vs-true-labels table also confirms weak separation.
   For Hierarchical Clustering, although it achieves a higher Silhouette Score of 0.66, the cluster-vs-true-labels comparison shows that almost all data points end up in a single cluster, meaning it does not reflect the true class structure.

   In the second dataset, both SVM models (Linear and RBF kernels) give moderate results. Accuracy is around 0.84**,** F1 is about 0.64**,** and AUC-ROC is high 0.89, but Precision and Recall are not very high.
   K-Means performs very poorly, with a Silhouette Score of 0.06, which is very low. The clusters vs true labels table shows a lot of mixing, so the clusters do not represent the real classes.
   Hierarchical Clustering, on the other hand, has a very high Silhouette Score 0.93, meaning the clusters are tight and well separated internally, but almost all data points end up in one cluster when compared to the true labels, so it does not match the actual classes

2. Which method seems to **separate the classes better**, and why?
   In the first dataset, **SVM** separates the classes better because it uses the true labels during training and therefore learns the optimal decision boundary. This results in high ROC, accuracy, and F1 scores.
   On the other hand, K-Means relies only on distance to form clusters, which is not effective for this dataset.
   Although Hierarchical Clustering achieves a high Silhouette Score, this metric only measures compactness and separation within the clustering itself and does not guarantee that the clusters correspond to the true labels.

   In the second dataset, **SVM** separates the classes better than K-Means because it uses the true labels to learn, which gives better results.
   K-Means fails to separate the data because it only considers distances, and the clusters are heavily mixed.
   Hierarchical Clustering forms very tight clusters (high Silhouette), but these clusters do not correspond to the true classes, so it cannot correctly separate the income classes.

3. How did **preprocessing** (encoding, scaling) help your models?
   Preprocessing helps the models learn well, as it is a necessary step. Encoding is important because the models can't work with categorical features. Scaling is important because all models rely on distances, so making all columns on the same scale prevents features with large values from dominating the calculations.

4. For each dataset, which method (SVM, K-Means, Hierarchical) would you **recommend**, and why?
   In the first dataset, I recommend using **SVM** because it achieves the highest and most reliable performance across all evaluation metrics.
   K-Means and Hierarchical Clustering may be useful for unsupervised analysis, but they do not align well with the true class labels, making them less suitable for classification in this dataset.

   In the second dataset, I would still recommend using **SVM** if we want to classify the data because it gives the best match to the true labels.
   K-Means is not recommended because it does not capture the class structure at all.
   Hierarchical Clustering could be useful for exploring the data structure, but it cannot be used as a reliable classifier because the clusters do not align with the true labels.

5. References

- ChatGPT
- Files in the BB